José Fabián Sifuentes Gálvez

Electronic Designer | Embedded Systems | RF & Satellite Communications

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Profile

Mechatronics Engineer (PUCP) specialized in electronic design and embedded systems, with 3+ years of experience in multilayer PCBs, C/C++ firmware on STM32 and Raspberry Pi, and RF communications for satellite missions. Experienced across the full hardware cycle: schematic capture, routing, physical prototype validation, sensor and peripheral integration, and UHF link testing. Proven track record in space projects (CubeSats, ground stations) and commercial products.

Experience

Professional Intern — Space Engineering · INRAS – PUCP · pre-professional until 2025 — Jul 2024 – Present

- Developed UHF communication systems for satellite missions, including link testing.
- Designed and developed the INTISAT ground station interface with real-time telemetry.
- Built 3D simulations of space tethers and integrated data link protocols.

Developer — Embedded Systems & Interfaces · Bear Tech — Jan – Jun 2025

- Internal pipeline-inspection robot: integration of 6 cameras (4 in a 360° array) on Raspberry Pi, IMU + odometry navigation (STM32, UART), and TCP/IP teleoperation.
- Automatic structural-fault detection with YOLOv8 integrated into the operator HUD (PyQt5).
- Designed the robot's electronic circuitry: component selection and integration of power and control stages.
- Designed multilayer PCBs with EasyEDA: schematics, routing, and physical prototype validation.

Student Researcher — Bionics · LIBRA – PUCP — Feb – Sep 2024

- Technical analysis of upper-limb myoelectric prostheses (electronics and actuation).
- OCTAHAND project: prosthesis recovery, repair, and improvement.

Fullstack Developer · Arborat · arborat.com · freelance — 2022 – Present

- Project management web application (React/Node.js on AWS) in production — full hardware-software perspective.

Featured Projects

AYNISAT — Peruvian CubeSat (CONIDA) · INRAS — 2024 – Present

- Communications payload with STM32: complete electronic design in EasyEDA and UHF link testing.

S-Band Transceiver — RF Design · INRAS — 2025 – Present

- RF chain design: amplifier + FPGA, DAC filter, modulator, clock PLL, TCXO, and patch antenna.
- Link budget analysis and justified component selection for the system.

INTISAT — UHF Communications Payload Lead · APSCO competition — 2024 – Present

- Telemetry system, satellite communications, and ground control station.

Digital PID Servo Control System · academic project, PUCP — 2025

- PID controller design under voltage constraints: MATLAB/Simulink synthesis and embedded deployment (PlatformIO) validated with real trajectories.

Technical Skills

Electronic Design: EasyEDA, KiCad · multilayer PCBs: schematics, routing, DRC, prototype validation
Embedded: STM32, Raspberry Pi 5, Arduino, FPGA (integration) · C/C++ firmware · PlatformIO
Buses & Protocols: I2C, SPI, UART, CAN · satellite telemetry, UHF links
RF & Communications: link budget, amplifiers, filters, PLL, TCXO, patch antennas, modulation
Control & Simulation: MATLAB/Simulink, PID design, ROS2, Gazebo, CoppeliaSim · robot kinematics
Software: Python, C, C++, MATLAB · PyQt5, Git, Linux/Windows
Mechanical Design: SolidWorks, Fusion 360, Autodesk Inventor, ANSYS

Education, Languages & Certifications

Mechatronics Engineering · Pontificia Universidad Católica del Perú (PUCP) — 2020 – 2025 · Lima, Peru
Languages: Spanish (native) · English — TOEFL (ICPNA)
Certifications: PERUMEC 2025 — robotics competitor · UNI Space TechWeek — speaker (LINKU project)